

Student's Name: _____

This exam contains **12** questions for the total of **20** points and is **9** pages long (including the cover). The number of points you get for answering correctly is given in **the left margin** of each question. To score **100%** you have to obtain **20** points.

You are allotted **45 minutes** to complete this exam.

In **Section 1** you need to mark the only correct option. This will be mostly questions like "true/false". This section will cost **30%** of the Exam.

In **Section 2** you need to mark the correct option or options. You need to **get the answer exactly correct** for **the full grade**: you get **zero** points for the entire question if you have irrelevant or missing options in your answer. This section will cost **30%** of the Exam.

**After completing these two sections put
your answers into the answer table on page 4.**

ONLY ANSWERS IN THE ANSWER SHEET WILL BE GRADED

In **Section 3** you are mostly required to give a full answer and demonstrate all relevant formulae and calculations. Write your answer and solution in the dedicated space below the question. You can use the last pages of the exam paper as draft pages. **Absent or illegible** answers will be graded zero. This section will cost **40%** of the Exam.

EXAMINATION RULES

- Students are required to follow all instructions given by the examiners.
- Talking is NOT allowed under any circumstances.
- Students should use the exam paper as a draft and a paper for solutions. The usage of any additional paper is prohibited.
- **Mobile phones are strictly prohibited** in the examination hall. Students MAY NOT bring any electronic device into the examination hall.
- Students may raise their hand to ask the examiner a question. The examiner may decide not to answer the question: the students are expected to know the required terminology and understand the examination questions.
- Once a student has seen the examination paper, the student is assumed to be in good health at the time of examination.
- Students are allowed to leave the examination room ONLY after they complete the exam.
- Students can complete the exam ahead of time. Early completion is not allowed within 20 minutes before the end of the exam.

I have read and understood the examination rules.
I will not cheat, copy from other students, or use unauthorized
materials or devices, and I have not brought such materials or
devices into the examination hall.

Signature: _____

Section 1

In this section, you should choose one of the two options. This section will cost **30%** of the Exam grade.

- (1) 1. What is machine learning closer to?
A. Astrology **B. Alchemy**
- (1) 2. What kind of task is a customer income prediction?
A. Classification **B. Regression**
- (1) 3. Which regression has a closed form solution?
A. LASSO **B. Ridge**
- (1) 4. What can be the cause of classifier's overfitting?
A. Too much features B. Too few features
- (1) 5. What should we do with an average entropy to maximize Information Gain for attributes in Decision Tree splits?
A. Minimize B. Maximize
- (1) 6. Which method of feature selection can handle non-linear dependencies in the data:
A. F-test **B. Mutual Information**

Section 2

In this section, you should choose one or many options. This section will cost **30%** of the Exam grade.

- (2) 7. Which metric is appropriate to evaluate a model on a task, where it is critical not to miss any positive example (e.g. fraud detection)?
- A. Precision
 - B. Recall**
 - C. F_β -score, where $\beta > 1$
 - D. F_β -score, where $\beta < 1$
- (2) 8. Which of the following statements about AdaBoost are TRUE?
- A. AdaBoost assigns higher weights to incorrectly classified samples in each iteration**
 - B. The final prediction of AdaBoost is based on a weighted vote of weak learners**
 - C. AdaBoost is highly resistant to noise and outliers in the data
 - D. If a weak learner performs better than random guessing, AdaBoost can improve it**
- (2) 9. You have a design matrix $X \in \mathbb{R}^{n \times d}$ with $d = 100,000$ features and a vector $y \in \mathbb{R}^n$ of binary 0-1 labels. When you fit a logistic regression model to your design matrix, your test error is much worse than your training error. You suspect that many of the features are useless and are therefore causing overfitting. What are some ways to eliminate the useless features?
- A. Use ℓ_1 regularization.**
 - B. Use ℓ_2 regularization.
 - C. Iterate over features; check if removing feature i increases validation error; remove it if not.**
 - D. If the i th eigenvalue λ_i of the sample covariance matrix is 0, remove the i th feature/column.

ANSWER SHEET

Section 1

1	2	3	4	5	6

Section 2

7	8	9

Section 3

10. You constructed a binary classifier that produced the following confusion matrix on the validation dataset. The positive and negative classes are labelled by $+$ and $-$, respectively. **Write a formula and calculate** the following values:

		predicted	
		$+$	$-$
actual	$+$	25	10
	$-$	5	40

The confusion matrix for problem 10.

- (1) (a) Recall

- (1) (b) F_1 score

Solution: We need to calculate **recall** (precision is calculated for F_1):

$$\text{recall} = \frac{\text{tp}}{\text{tp} + \text{fn}} = \frac{25}{25 + 10} = \frac{5}{7}$$

$$\text{precision} = \frac{\text{tp}}{\text{tp} + \text{fp}} = \frac{25}{25 + 5} = \frac{5}{6}$$

Formula for F_1 :

$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} = \frac{2}{\frac{7}{5} + \frac{6}{5}} = \frac{10}{13}$$

- (3) 11. Gradient Boosting libraries often have a regularization parameter called `min_samples_leaf` or something similar. Explain what it controls. How such a regularization works?

Solution: The `min_samples_leaf` parameter is a regularization parameter in Gradient Boosting that controls the minimum number of samples required to be in a leaf node of the decision trees used as weak learners. Without a minimum leaf size constraint, trees might create leaf nodes with very few samples, which are likely to capture noise or outliers in the training data. By requiring a minimum number of samples in each leaf, the algorithm ensures that the leaf nodes represent more generalizable patterns rather than spurious noise.

- (3) 12. Show that the ordinary least squares (OLS) loss function for linear regression is convex in w .

The OLS loss function is given by:

$$L(w) = \frac{1}{2m} \sum_{i=1}^m (y_i - w^\top x_i)^2.$$

Hint: Use matrix form.

Solution: The OLS loss function is given by:

$$L(w) = \frac{1}{2m} \sum_{i=1}^m (y_i - w^\top x_i)^2.$$

This can be rewritten in matrix form as:

$$L(w) = \frac{1}{2m} \|y - Xw\|^2,$$

where X is the design matrix with rows $x_i^\top$.

Expanding the squared norm:

$$L(w) = \frac{1}{2m} (y^\top y - 2y^\top Xw + w^\top X^\top Xw).$$

To analyze the convexity of $L(w)$, we compute its Hessian. First, we find the gradient:

$$\nabla L(w) = \frac{1}{2m} \cdot 2 (X^\top Xw - X^\top y) = \frac{1}{m} (X^\top Xw - X^\top y).$$

Now, differentiating again to obtain the Hessian:

$$H = \nabla^2 L(w) = \frac{1}{m} X^\top X.$$

Since $X^\top X$ is a Gram matrix, it is always positive semi-definite:

$$v^\top X^\top X v \geq 0, \quad \text{for any } v.$$

Hence, H is also positive semi-definite, proving that $L(w)$ is convex.

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